

CSI 500 Covered Call Strategy Analysis

Strategy introduction

Strategy logic

At the beginning of the backtest, buy 10,000 ETF units as the underlying position and simultaneously sell the corresponding number of front-month Call contracts at the selected strike level.

At each monthly option expiry, automatically check whether exercise occurs. If $S < K$, the Call expires worthless; retain the premium and sell the next front-month Call at the selected level. If $S > K$, buy 10,000 ETF units again after assignment and sell the next front-month Call at the selected level.

Sources of profit

From the trade perspective, each month has an exercised or unexercised outcome. If the Call is not exercised, profit comes from the premium received. If it is exercised, the underlying has risen above the strike relative to its purchase price. Assignment sells it at the strike, realizing $P_{\text{real}} = K - S_{\text{entry}}$, together with the option premium. However, assignment gives up the additional underlying gain above the strike: $P_{\text{market}} = S_{\text{current}} - K$.

From the Greeks perspective, the original report attributes profit to a small residual portfolio Delta, depending on the sold option's strike level, and to the rapid time-value decay of front-month Calls. It describes selling Calls as “short theta”; that terminology is retained here as an original research statement.

Sources of risk

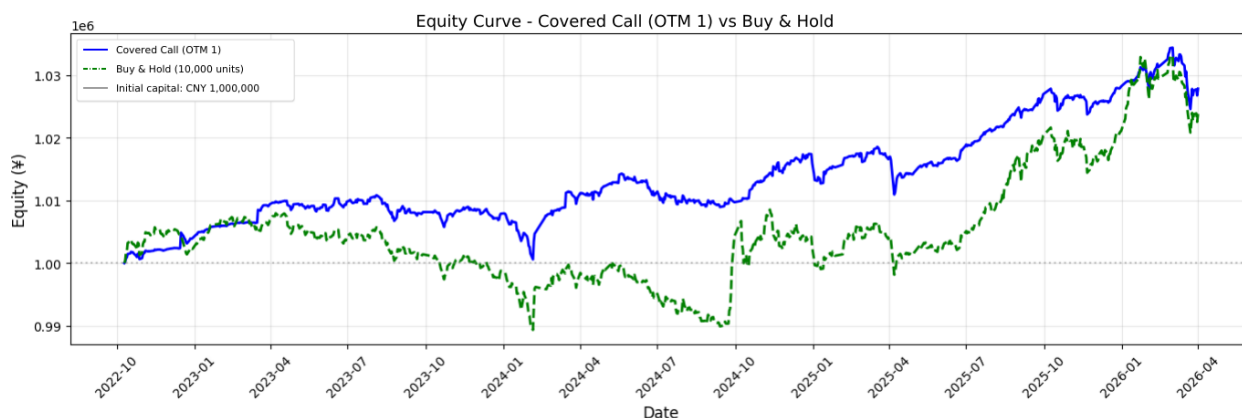
Selling Calls creates short Gamma exposure, so large underlying price moves can cause losses. It also creates short Vega exposure, which the original report identifies as another source of losses during large market moves.

Strategy backtest

The study uses daily CSI 500 ETF data from the second half of 2022 to the first half of 2026. Rebalancing follows the strategy logic described above.

The original parameter description specifies initial capital of CNY 1,000,000, an option price of CNY 0.8 per contract, an ETF commission of 0.03%, and option slippage of 0.1%. No ETF slippage was applied because the report considers the CSI 500 ETF liquid. It states that the low rebalance frequency limits the effect of fees and slippage on the backtest.

One-strike OTM Call results

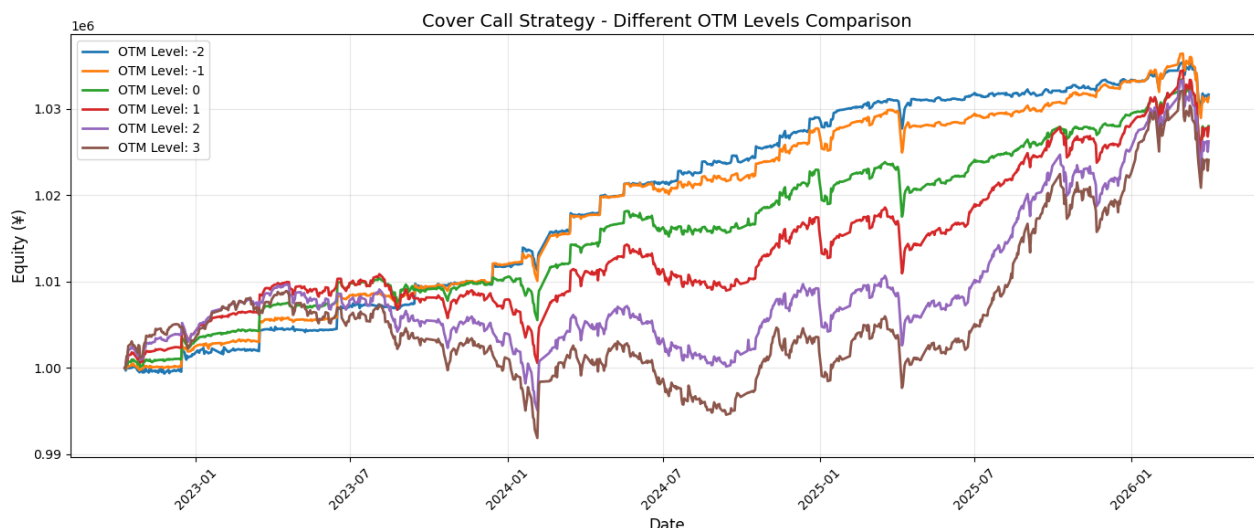


Original covered-call and long-only equity curves

The reported covered-call annualized return is 0.82% and maximum drawdown is 3.33%. The long-only benchmark returns 0.71% annually with maximum drawdown of 4.24% over the same period.

The report interprets premium income as providing a small return enhancement of about 0.11 percentage points annually and useful downside protection, narrowing maximum drawdown by almost one percentage point. It concludes that the strategy met its goal of enhancing return and smoothing the equity curve. Absolute return remained low, which it attributes to a range-bound or mild market in which the premium-income characteristic was evident.

Comparison across Call strike levels



Original covered-call comparison across strike levels

The original figure labels levels -2 and -1 as two and one strikes ITM, level 0 as ATM, and levels 1, 2, and 3 as one, two, and three strikes OTM. A typographical ambiguity in the original prose is clarified using these unchanged figure labels.

The report describes a clear performance gradient as the sold Call strike moves lower, from deep OTM toward ITM. OTM1 outperforms OTM2 and OTM3; ATM improves return further; ITM achieves the highest return. It characterizes the results as progressively better performance at lower, more deeply ITM strikes.

Two explanations are offered. First, lower strikes provide more downside protection. A more deeply ITM sold Call contains more intrinsic value in its premium, creating a thicker cushion. In the study's range-bound or mild market, this cushion offsets small underlying declines, supports positive returns on more days, controls drawdown, and improves cumulative return.

Second, the market environment favors ITM covered calls. This structure gives up substantial upside participation and shifts the exposure from directional investing toward premium income. Its lower Delta smooths NAV in sideways or weak markets and avoids some losses experienced by a long-only position during large fluctuations. In a strong bull market, the capped upside could substantially underperform OTM strategies. The report states that its sample did not include a sustained explosive rally, allowing the ITM structure's advantages to appear.

The report stresses that this relationship is not universal and depends on market conditions. OTM Calls retain more upside in a sustained rising market; ITM Calls offer larger premium cushions in sideways or falling markets. It therefore proposes adapting strike levels to an expected market regime rather than always using a fixed level.

Live liquidity considerations

Option contracts are less liquid than the underlying ETF. Deeply OTM, near-expiry, or less actively traded months can have wide bid-ask spreads and difficulty obtaining fills. Under extreme conditions, liquidity can

deteriorate further, preventing position adjustments at reasonable prices. The report recommends prioritizing active contract months and avoiding deeply OTM or obscure contracts to reduce liquidity discounts.

Proposed strategy improvements

Direction 1: dynamic entry strike selection

The proposed improvement is to adjust the strike of the sold Call according to the expected market regime. When a sustained upward trend is expected, sell OTM Calls to retain much of the underlying's upside and avoid capping gains too early. When sideways or declining conditions are expected, switch to ITM Calls, using larger premiums as a cushion against downside while enhancing holding returns.

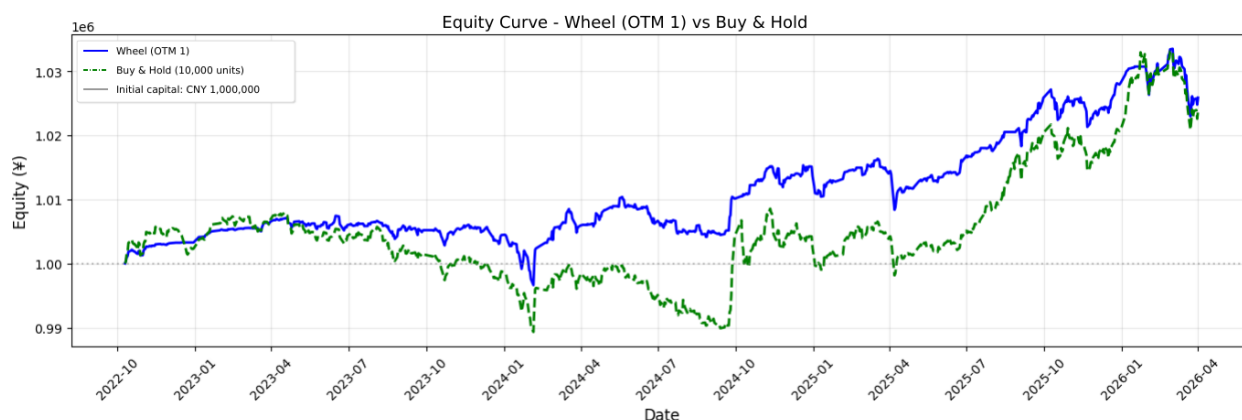
The report describes this flexible framework as selling OTM in bull markets and ITM in bear or sideways markets, seeking a changing balance between return enhancement and risk controls across market regimes.

Direction 2: Covered Call to Wheel

A covered-call strategy enhances return through premiums but requires continued ownership of the underlying. Entry timing is passive: if the ETF is purchased at a high price and then falls persistently, premiums may not offset large losses. Strong upward trends also face a cap on gains.

The report proposes a Wheel strategy to address these limitations. First sell Puts to earn premiums while waiting for a pullback to a desired entry price, seeking discounted entry. After acquiring the ETF, sell Calls for continued premium income. This creates a recurring cash-to-holdings-to-cash cycle. The report describes switching between Put selling and Call selling as improving timing and risk management compared with the basic covered-call structure.

A simple Wheel backtest was also conducted in the supplied system.



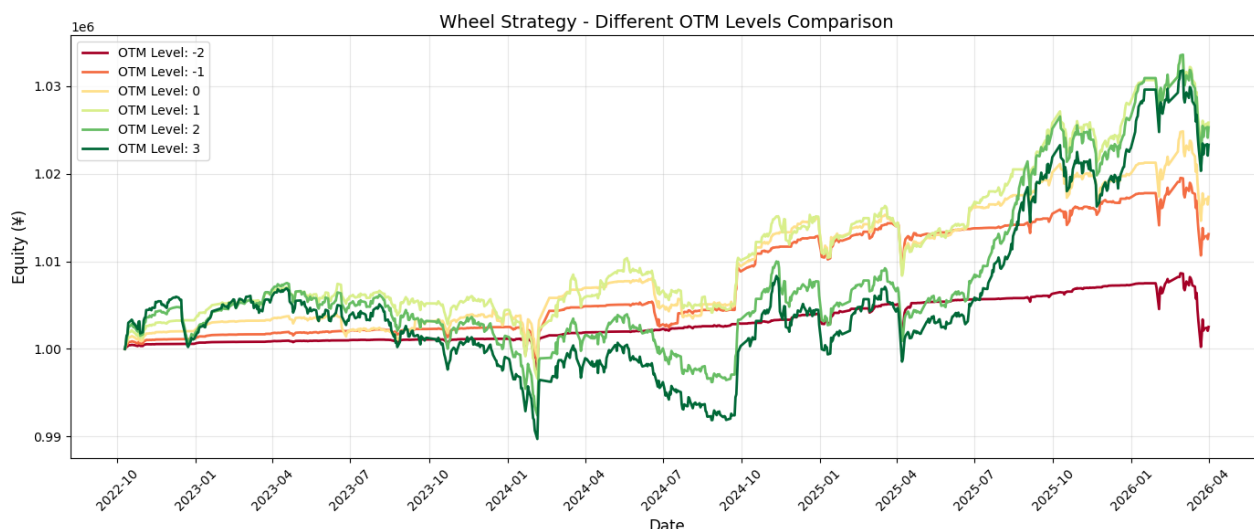
Original Wheel and long-only equity curves

The one-strike OTM Wheel has a reported annualized return of 0.76% and maximum drawdown of 3.57%. Its return is slightly below the covered call's 0.82%, and its drawdown is slightly larger than 3.33%, giving it weaker overall performance in this sample.

Both strategies improve return and narrow drawdown compared with the long-only benchmark. In the study's largely range-bound period, the simple always-hold-the-ETF-and-sell-Calls structure performs slightly better. The Wheel enters through Put selling and can wait longer without holding the ETF, so it does not fully realize the benefit of discounted entry in a mild upward market. The report attributes its slightly weaker return and risk metrics to this effect.

The report considers the Wheel better suited to sideways or declining markets, where receiving premiums before entering at a lower price can be more useful. During a mild rise, continuous ETF ownership in the simpler covered-call structure can be advantageous, explaining the comparison observed in this sample.

Wheel strike-level comparison



Original Wheel comparison across strike levels

From October 2022 to April 2026, the Wheel equity curves differ noticeably across strike levels. Deep ITM levels are nearly flat with almost no return; mildly ITM positions are also stable but have limited gains. The ATM curve rises steadily and performs clearly better than ITM positions. OTM curves largely overlap, ending at similar returns, but all below ATM.

The report concludes that ATM options achieved the best balance between return enhancement and risk controls over this period. It attributes limited ITM returns to a large intrinsic-value component and little time value, and limited OTM protection to thin premiums. The reported ordering is ATM above OTM, and OTM above ITM.

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